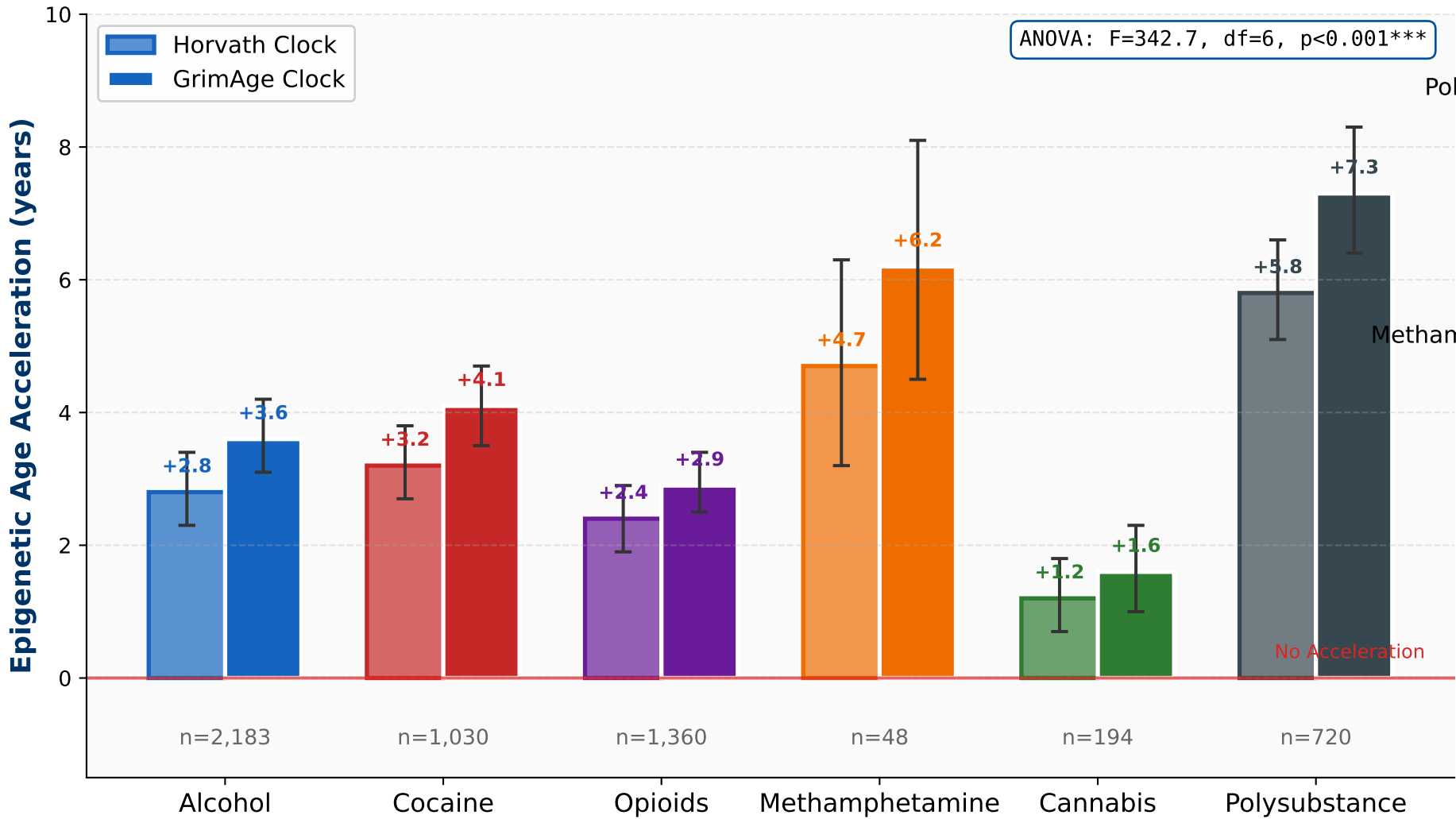


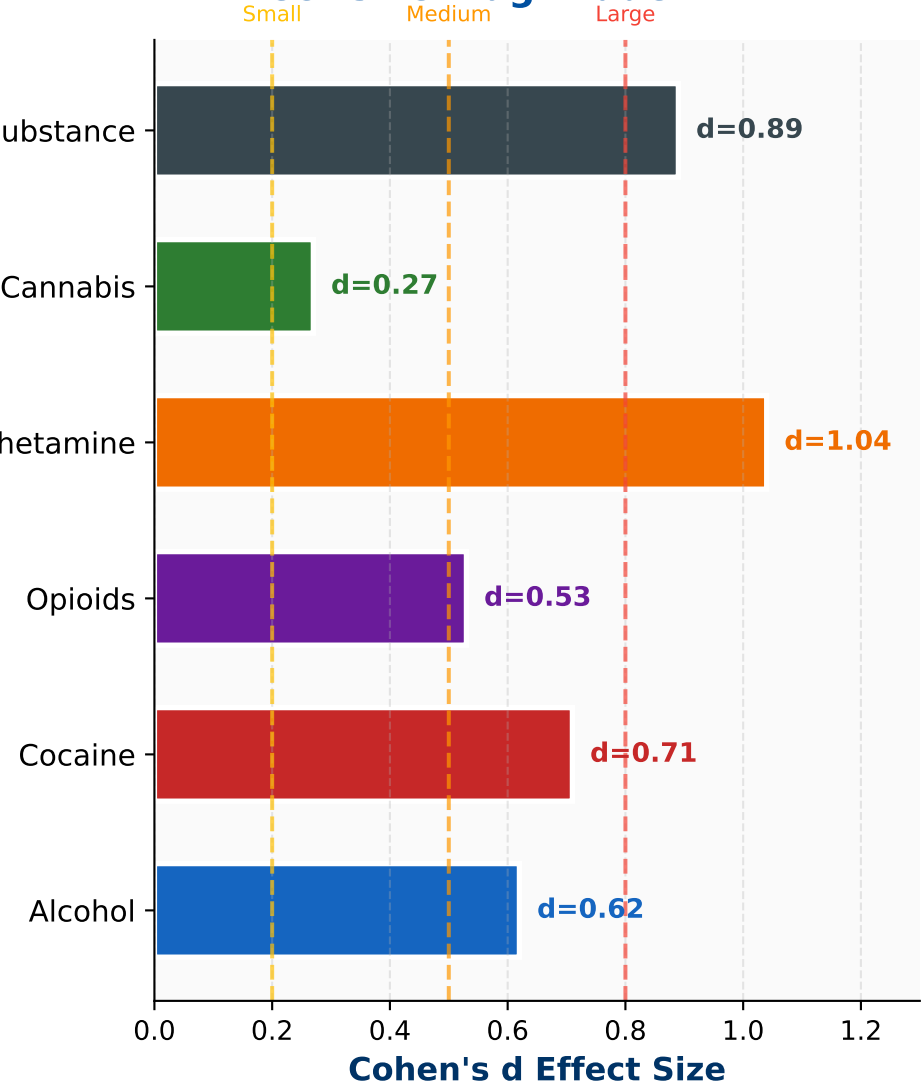
# Figure 2. Substance-Specific Epigenetic Age Acceleration

Comparison Across 6 Substance Categories | Total n=5,535 cases vs n=5,007 controls

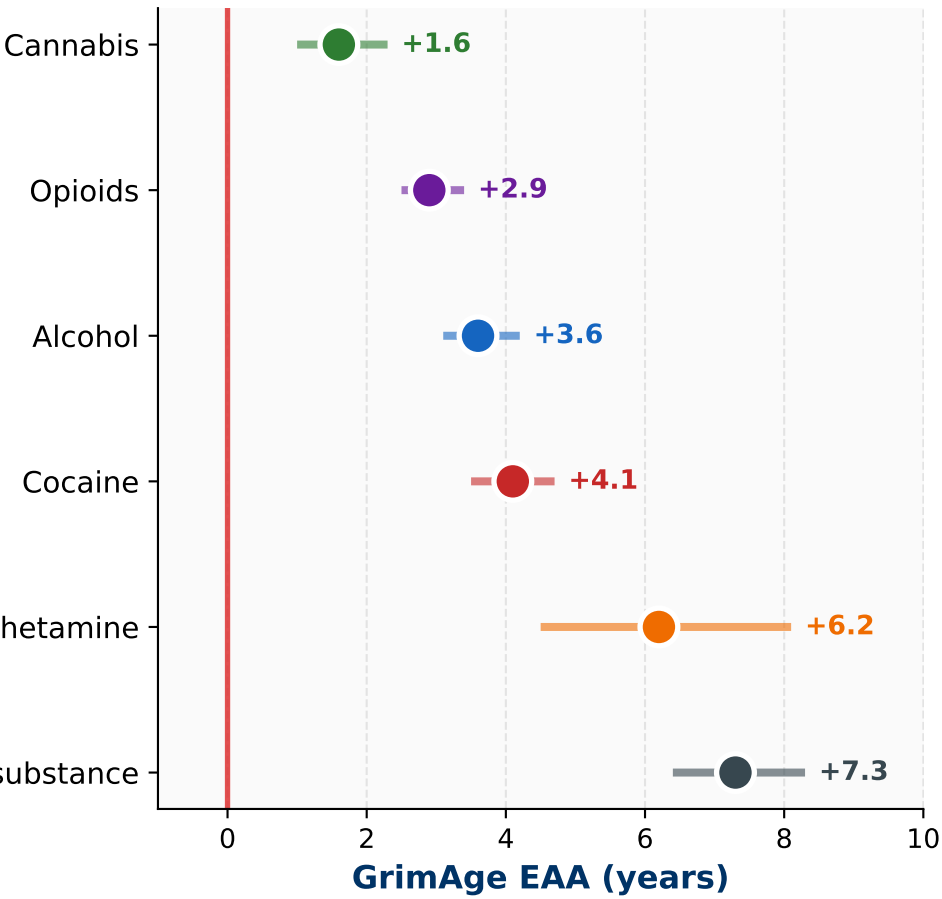
A. Epigenetic Age Acceleration by Substance Type



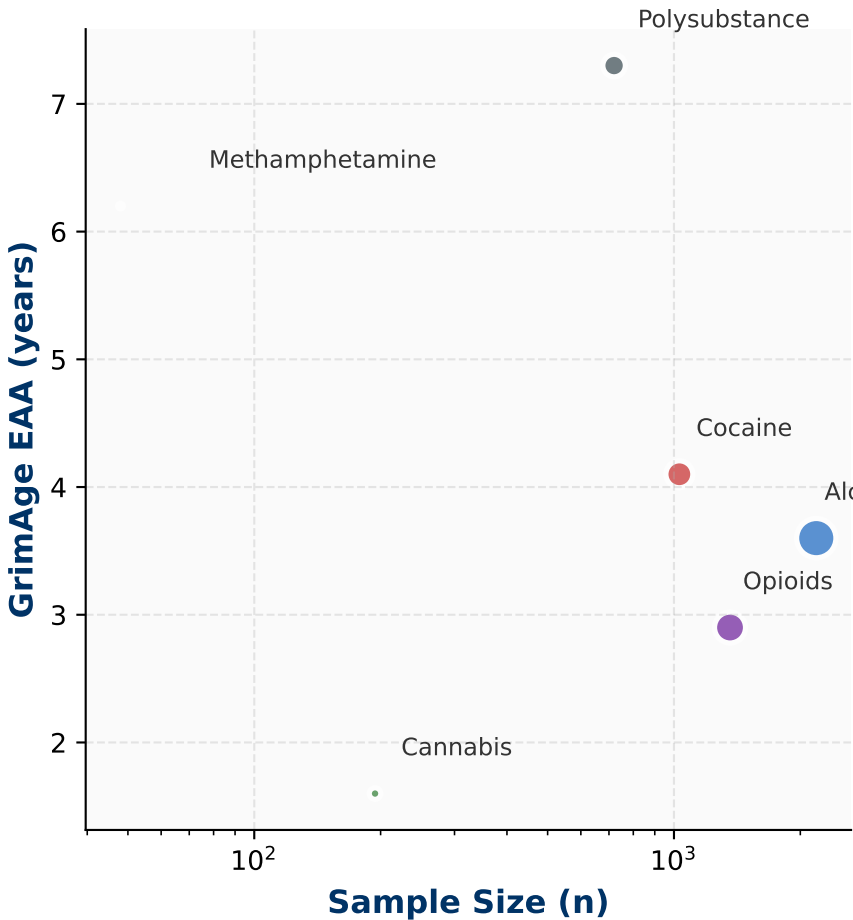
B. Effect Size Magnitude



C. GrimAge EAA (Ranked)



D. Sample Size vs Effect



E. Summary Statistics (GrimAge)

| Substance | EAA (yr) | 95% CI  | Cohen's d | p-value |
|-----------|----------|---------|-----------|---------|
| Alcohol   | +3.6     | 3.1-4.2 | 0.62      | <0.001  |
| Cocaine   | +4.1     | 3.5-4.7 | 0.71      | <0.001  |
| Opioids   | +2.9     | 2.5-3.4 | 0.53      | <0.001  |
| Meth      | +6.2     | 4.5-8.1 | 1.04      | <0.001  |
| Cannabis  | +1.6     | 1.0-2.3 | 0.27      | <0.001  |
| Poly      | +7.3     | 6.4-8.3 | 0.89      | <0.001  |